

Supabase Vs. Firebase

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Alternative

Supabase vs Firebase

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Firestore and Supabase both promise to give teams a full-featured backend without managing servers. Firestore, owned by Google, combines a NoSQL database with authentication, file storage and serverless functions. Supabase is an open-source alternative built on Postgres that offers similar services—auth, storage, real-time, and functions, but with the predictability of SQL and the freedom to self-host. Choosing between them depends on how your application models data, how you plan to scale, and your tolerance for vendor lock-in.

What is Supabase?

Supabase is an open-source backend platform that runs on Postgres. Each project comes with a dedicated Postgres database, an auto-generated REST and GraphQL API, real-time subscriptions, and storage, all tied together with Row-Level Security (RLS). RLS policies are written in SQL and allow you to define granular access rules using the same language you use for your data. Because Supabase's core services are built on open technologies like Postgres and GoTrue, you can run Supabase locally or host it yourself using Docker or community tools such as Kubernetes or Terraform.

What is Firebase?

Firebase is a managed platform owned by Google. It includes Cloud Firestore (a document-oriented NoSQL database), authentication, Cloud Storage for files and Cloud Functions for backend logic. Firestore stores data as JSON-like documents grouped in collections. The SDKs automatically cache documents on the device so apps can read, write and listen to data while offline; changes are synchronized when the device reconnects. Firestore supports collection group queries, which let you query across collections with the same name. To handle relational data you often denormalize documents or perform multiple queries in code.

Conclusion

Firebase offers an excellent developer experience for prototypes and simple applications. It excels at real-time synchronization and has a generous free tier. Supabase provides a similar developer experience with a relational core. By building on Postgres and open-source components, Supabase gives you SQL queries, ACID transactions, RLS and the option to self-host. For teams that anticipate complex data relationships, need fine-grained access control, or want predictable costs and open-source flexibility, Supabase is a compelling alternative.

Need help migrating from Firebase to Supabase? [Contact](#) our Firebase Migration Team.

Core architecture and database

Feature	Firebase	Supabase
Database type	Document database (Cloud Firestore)	Relational database (Postgres)
Data model	Schemaless JSON; subcollections; no native joins	Structured tables with foreign keys and indexes
Transactions	Supports transactions and batched writes	Full ACID transactions built into Postgres; you can execute complex joins, subqueries and data transformations in SQL
Offline support	Firestore caches actively used data on the client so you can read/write while offline; changes sync when connectivity returns	Supabase clients use Postgres directly; offline access requires your own caching strategy and works with Zero, Electric,
API access	Client SDKs for web, Android, iOS and server environments; REST/gRPC endpoints	Auto-generated REST API via PostgREST and GraphQL API via pg_graphql
Self-hosting	Not supported; services run on Google Cloud	Fully self-hostable via Docker or on your own cloud

Firestore's schemaless design makes it easy to prototype. As data relationships grow more complex, you must handle joins in the client or denormalize your data. Supabase's Postgres foundation gives you SQL's expressive power, foreign keys, and indexes from day one.

Authentication and user management

Feature	Firebase	Supabase
Login methods	Email/password, phone, anonymous and OAuth providers; additional SAML and OpenID Connect support requires upgrading to <i>Firebase Authentication with Identity Platform</i>	Email/password, OAuth providers (Google, GitHub, Apple, etc.), SMS and custom providers
Enterprise auth	The Identity Platform upgrade enables multi-factor auth, SAML and OIDC	SSO, SAML and OIDC are available on Pro and Enterprise plans without changing SDKs
Access control	Security Rules use a domain-specific language and differ for Firestore, Storage and Functions	Row-Level Security (RLS) policies written in SQL control access across the database
Offline auth	The client SDK persists auth tokens locally and handles session renewal	Session management integrates with Postgres; tokens are validated on the server
Customization	Custom auth requires writing Cloud Functions or backend services	You can write custom policies and logic in SQL; column-level security enables field-level restrictions

Firebase Authentication is easy to set up for email/password and social logins. Enterprise features such as SAML/OIDC and blocking functions require upgrading to the Identity Platform and come with new usage limits. Supabase Auth integrates with Postgres, so you can enforce complex policies using SQL and upgrade to SSO or enterprise auth features without changing products.

Serverless functions and backend logic

Feature	Firebase	Supabase
Runtime	Cloud Functions run on Google's infrastructure. You write code in JavaScript or TypeScript and deploy through the Firebase CLI; Python support is available in the second-generation runtime	Edge Functions run on Deno and are globally distributed at the edge for low-latency execution
Languages	JavaScript, TypeScript and Python (2nd gen)	TypeScript (via Deno); you can also call third-party APIs or write SQL from functions
Triggers	HTTP requests, Firestore, auth events, Storage, pub/sub and scheduled jobs	HTTP requests, scheduled jobs and other triggers; functions can call Postgres directly through the <code>supabase-js</code> client
Cold start	Functions run in a managed environment; cold starts can vary depending on region and load	Edge Functions start quickly because Deno's runtime is lightweight and runs close to users

Firebase Cloud Functions integrate tightly with other Firebase products and support a wide range of triggers. Supabase Edge Functions are TypeScript functions that run on Deno at edge locations. They talk directly to your Postgres database, which simplifies tasks like handling webhooks or building custom APIs.

Storage and file management

Feature	Firebase	Supabase
Underlying platform	Cloud Storage for Firebase (built on Google Cloud Storage)	S3-compatible object store integrated with Postgres
Resumable uploads	The web SDK provides <code>uploadBytesResumable()</code> and methods to pause, resume or cancel an upload	Built-in support for resumable uploads; each upload is recorded in a Postgres table
Access control	Security Rules use a separate rules language for Storage	Storage metadata is stored in Postgres and protected by the same RLS policies as the rest of your data
Image tools	No built-in transformations (you can add Firebase Extensions for resizing)	Built-in image transformations and CDN delivery
Free tier	Spark plan includes 5 GB of Cloud Storage in select regions; additional usage is billed by region and operation	Free plan allows files up to 50 MB and 1 GB total storage; Pro/Team plans allow up to 500 GB per file and include CDN

Firebase Storage is reliable and integrates with the auth SDK. Pricing depends on region and includes separate charges for storage, bandwidth and operations. Supabase Storage integrates with Postgres: metadata is stored in a table, so you can apply the same RLS policies you use elsewhere. Free plans include 50 MB per file, and Pro plans raise that limit to hundreds of gigabytes.